Assembly Guide
Laser Sprint

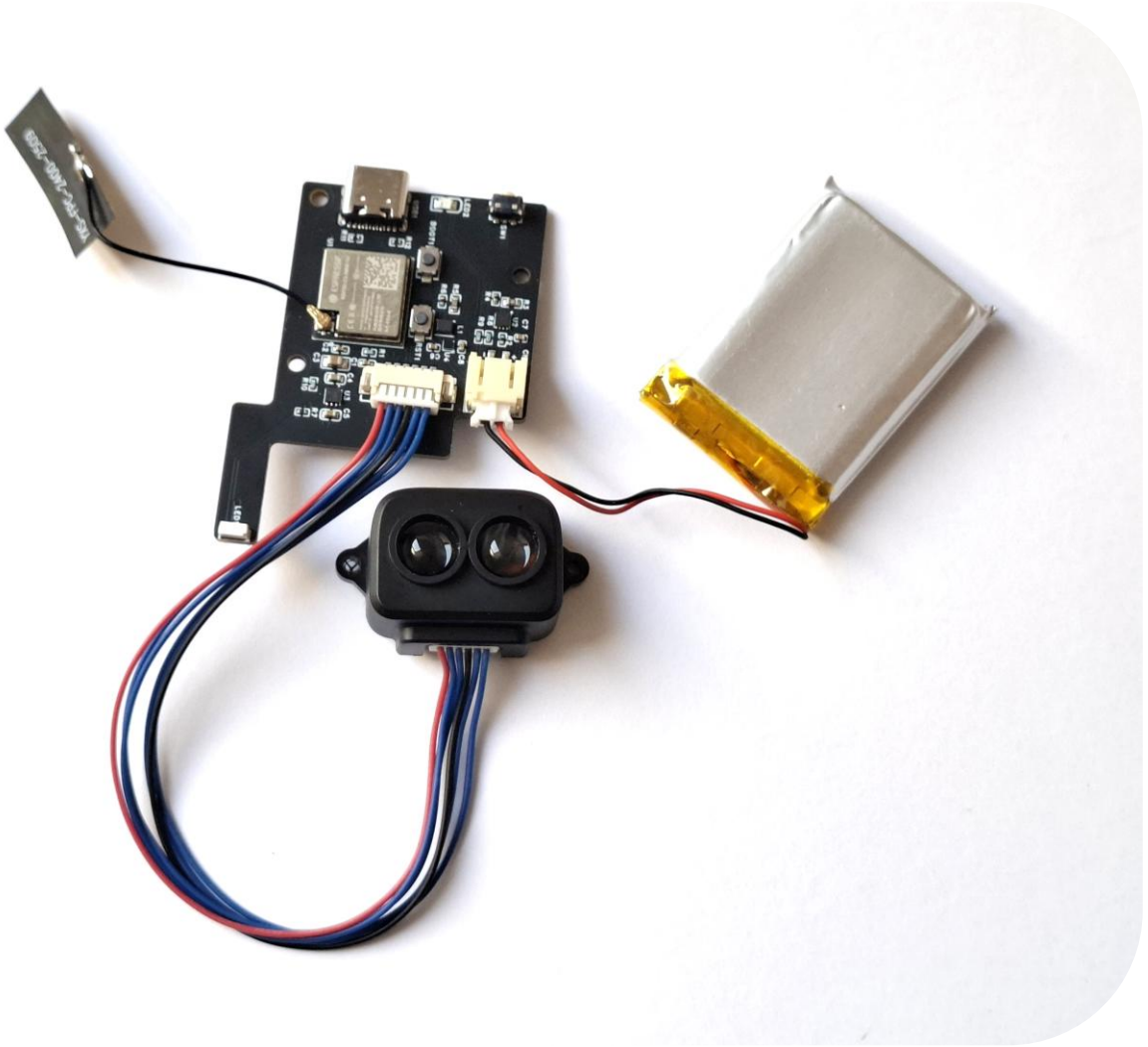


V1.0

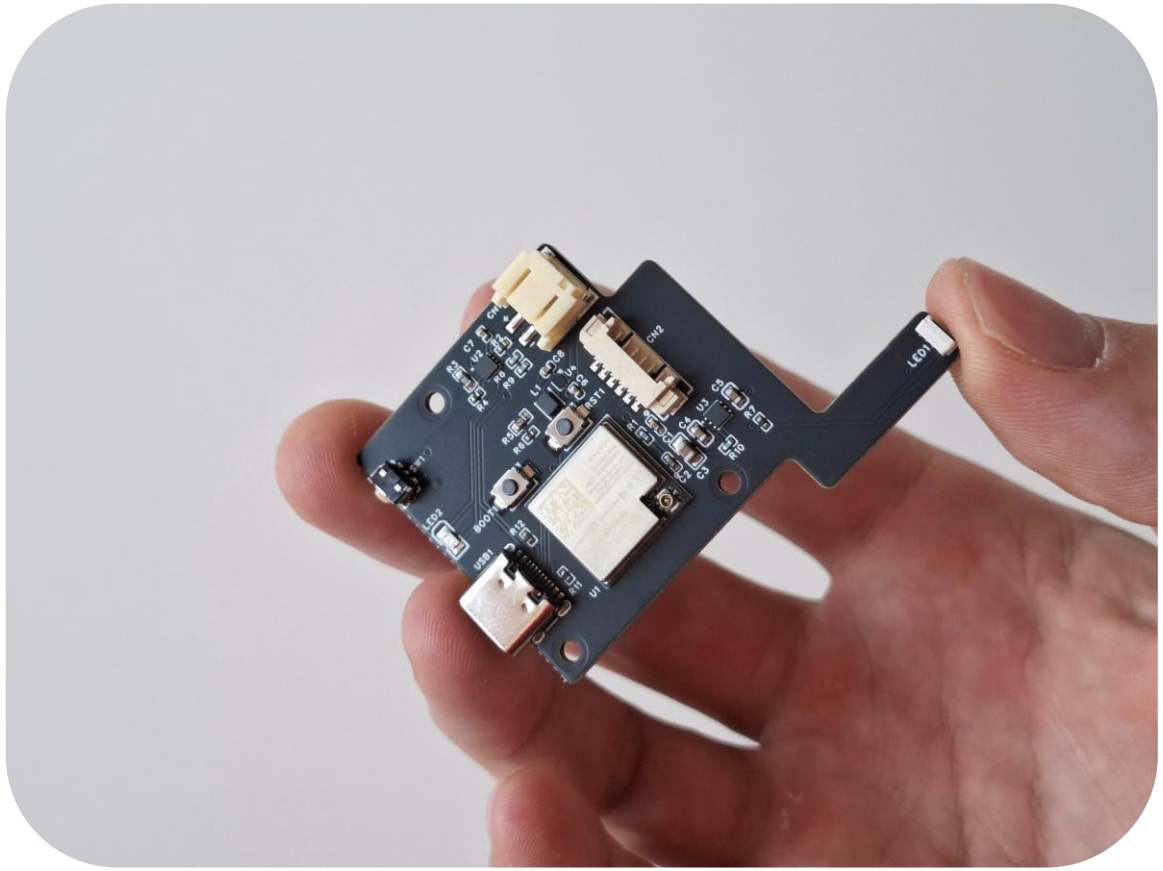
What accessories/tools will you need?

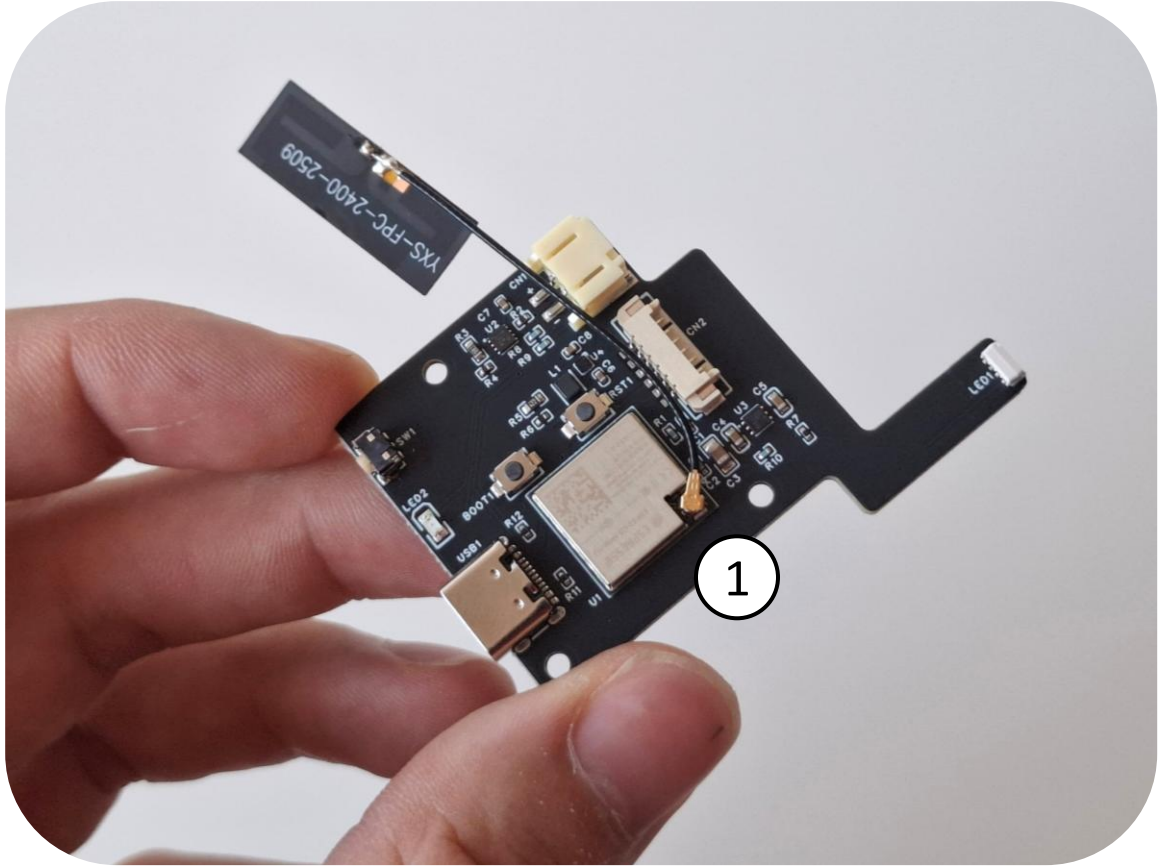
- **Custom PCB:**
  - Pre-assembled Laser Sprint board (PCBWay)
- **LiDAR Sensor:**
  - TF-Luna ToF LiDAR 8m (UART)
- **LiPo Battery:**
  - 3.7V 1000mAh (Size 803040 / ~40x30mm)
- **BLE Antenna:**
  - 2.4GHz 3dBi with **IPEX3** connector
- **Tripod Mount:**
  - 1/4"-20 UNC Threaded Brass Insert (Short)
- **Screws:**
  - M2x8mm Screws for 3D Printed Case
- **3D Printed Enclosure**
- 👉 **Download 3D model MakerWorld**



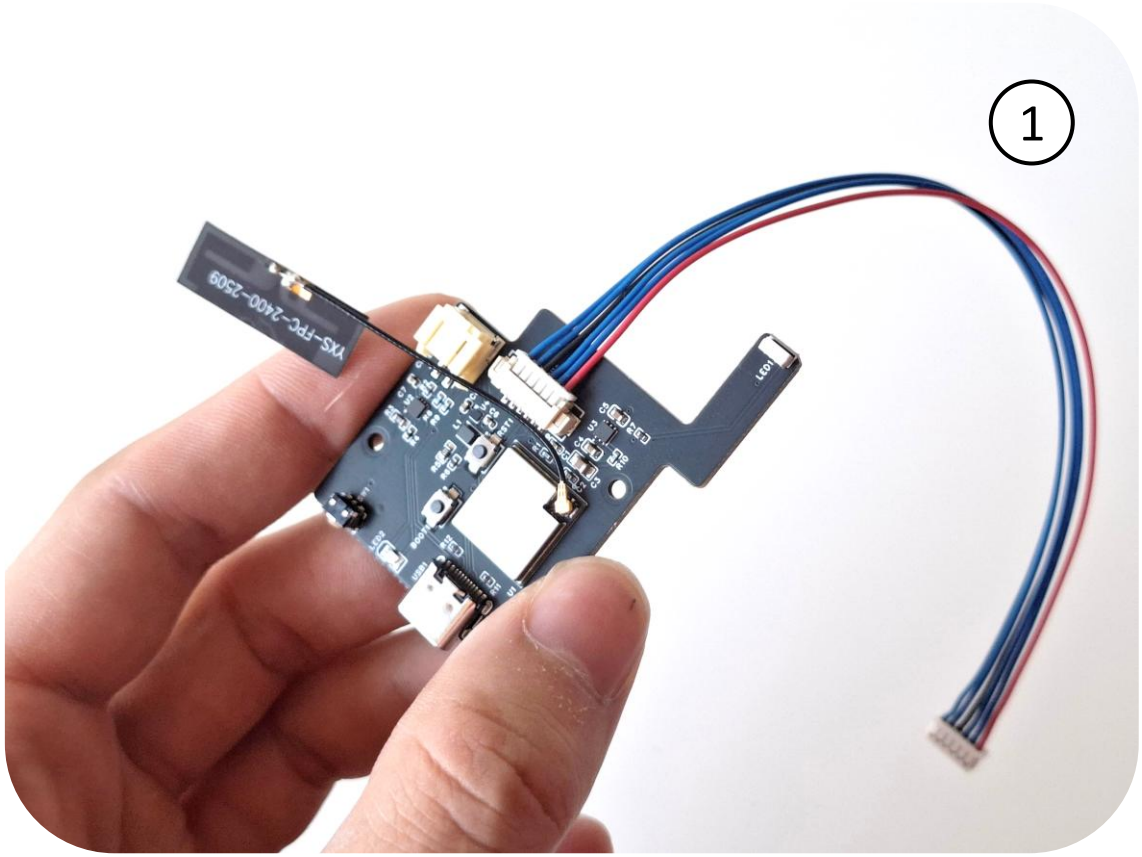


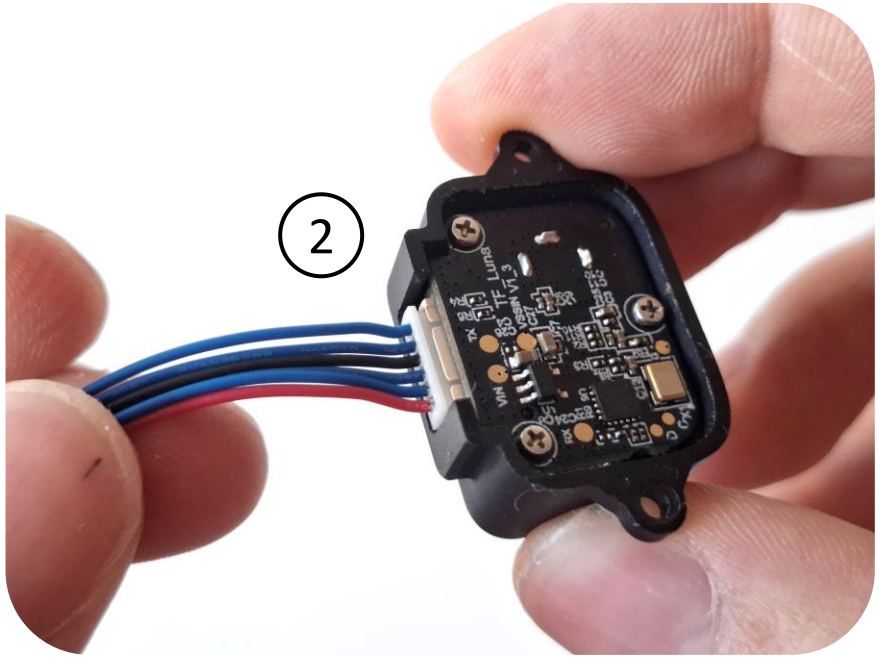
Let’s start!

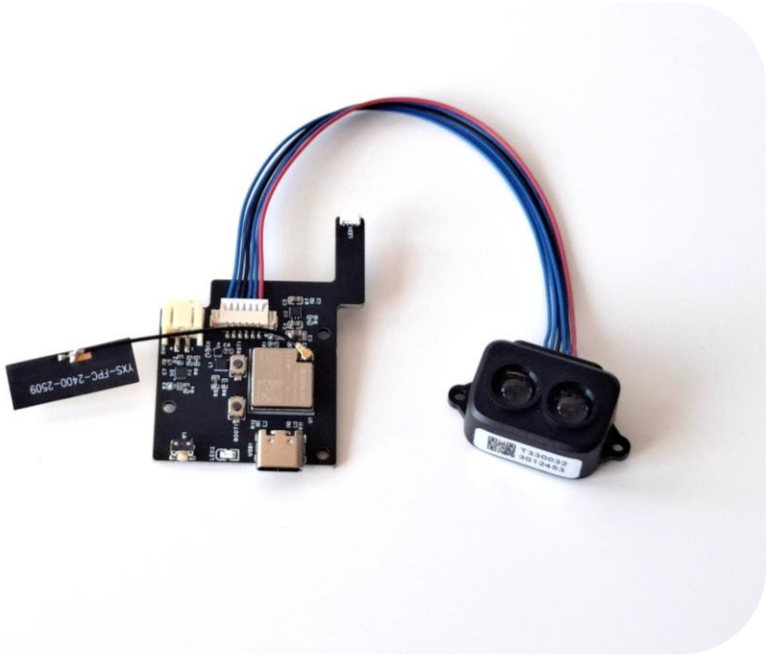




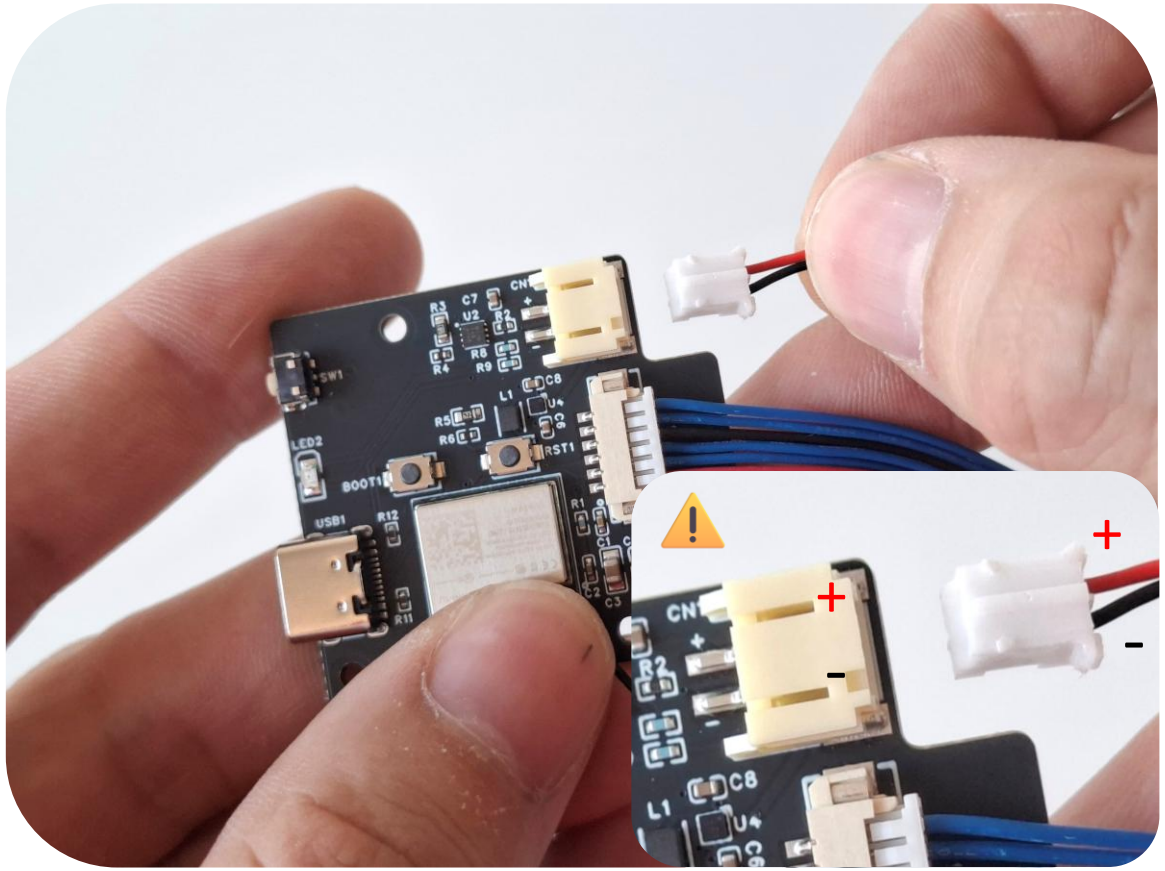
1 Carefully connect the antenna with an IPEX3 connector to the ESP32-C3 module.

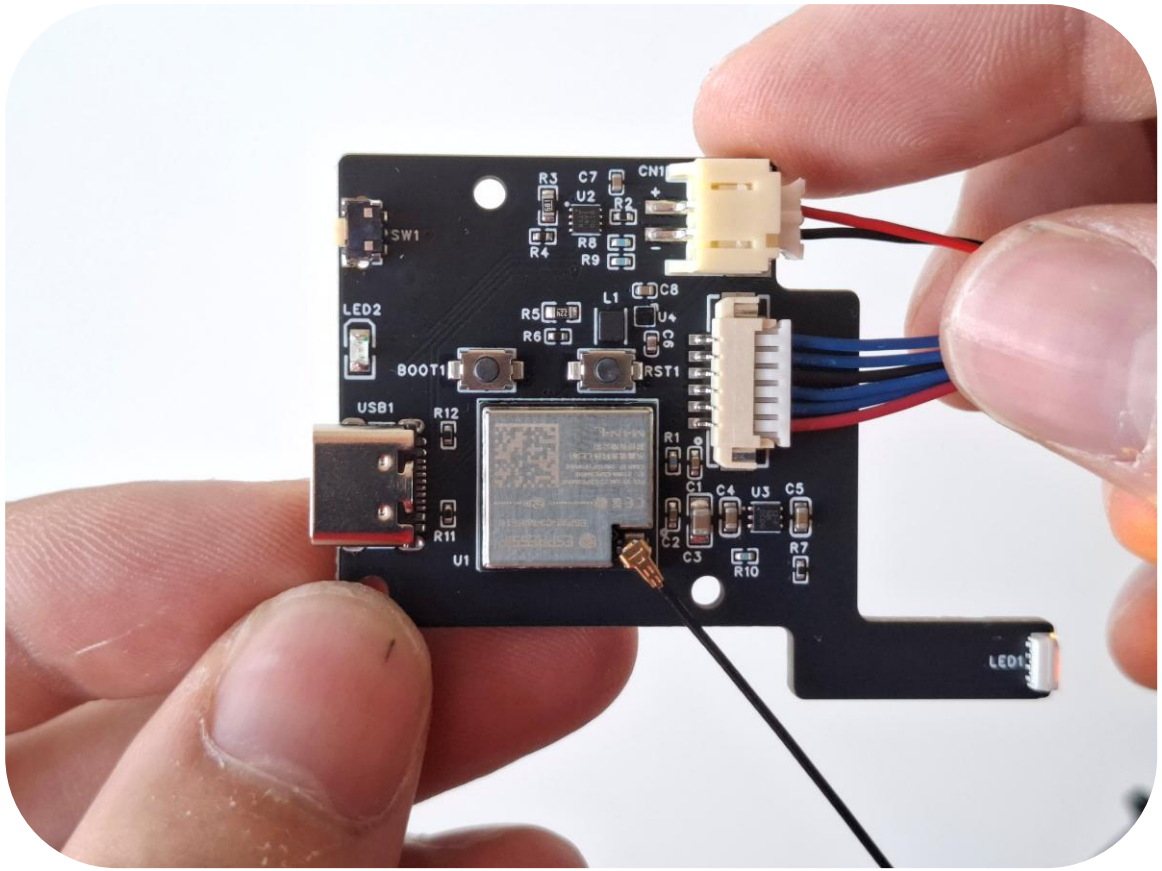






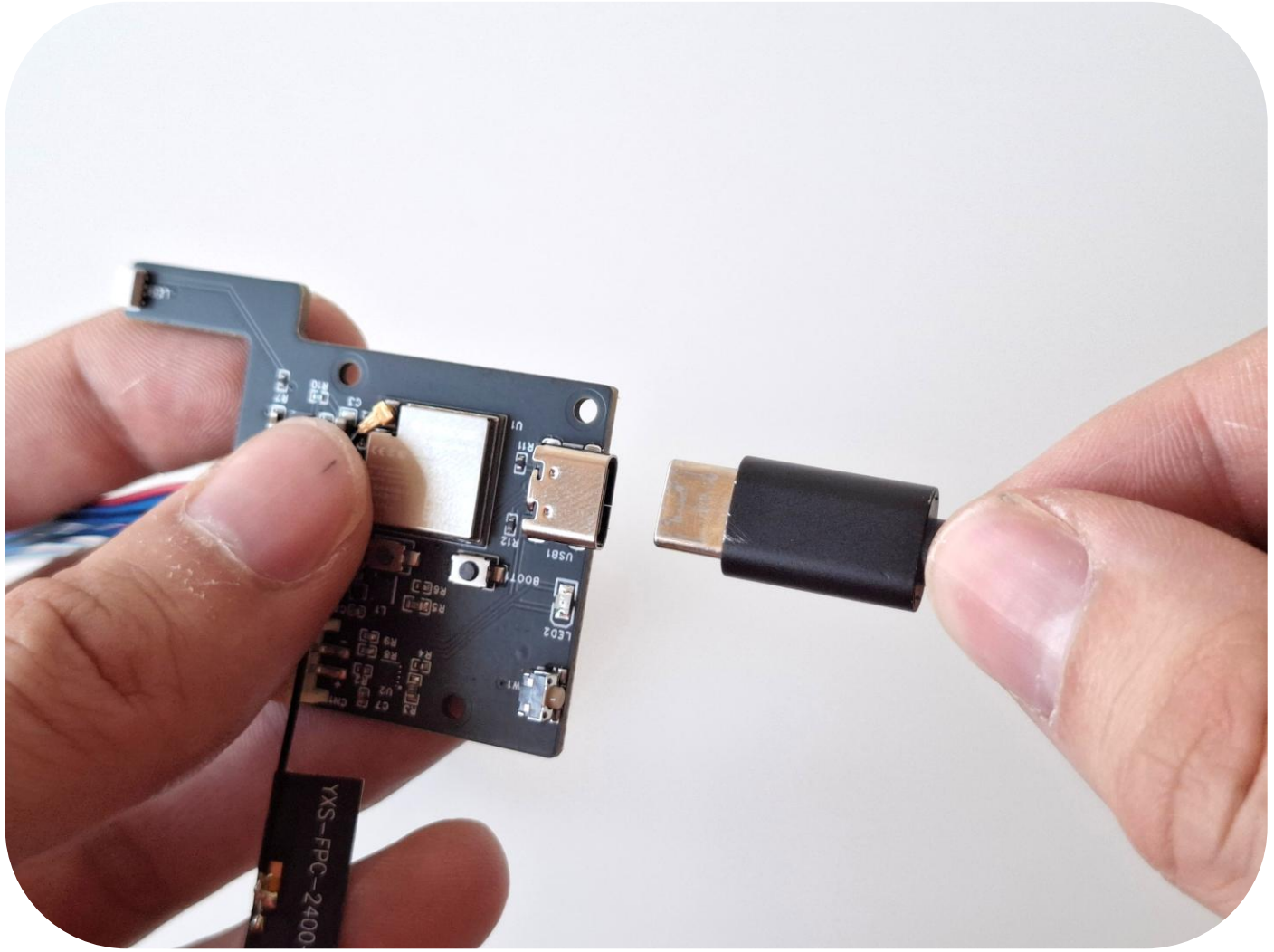
1. Connect the laser cable (included in the laser package).
2. Connect the laser module to the other end of the cable.





① Connect the battery

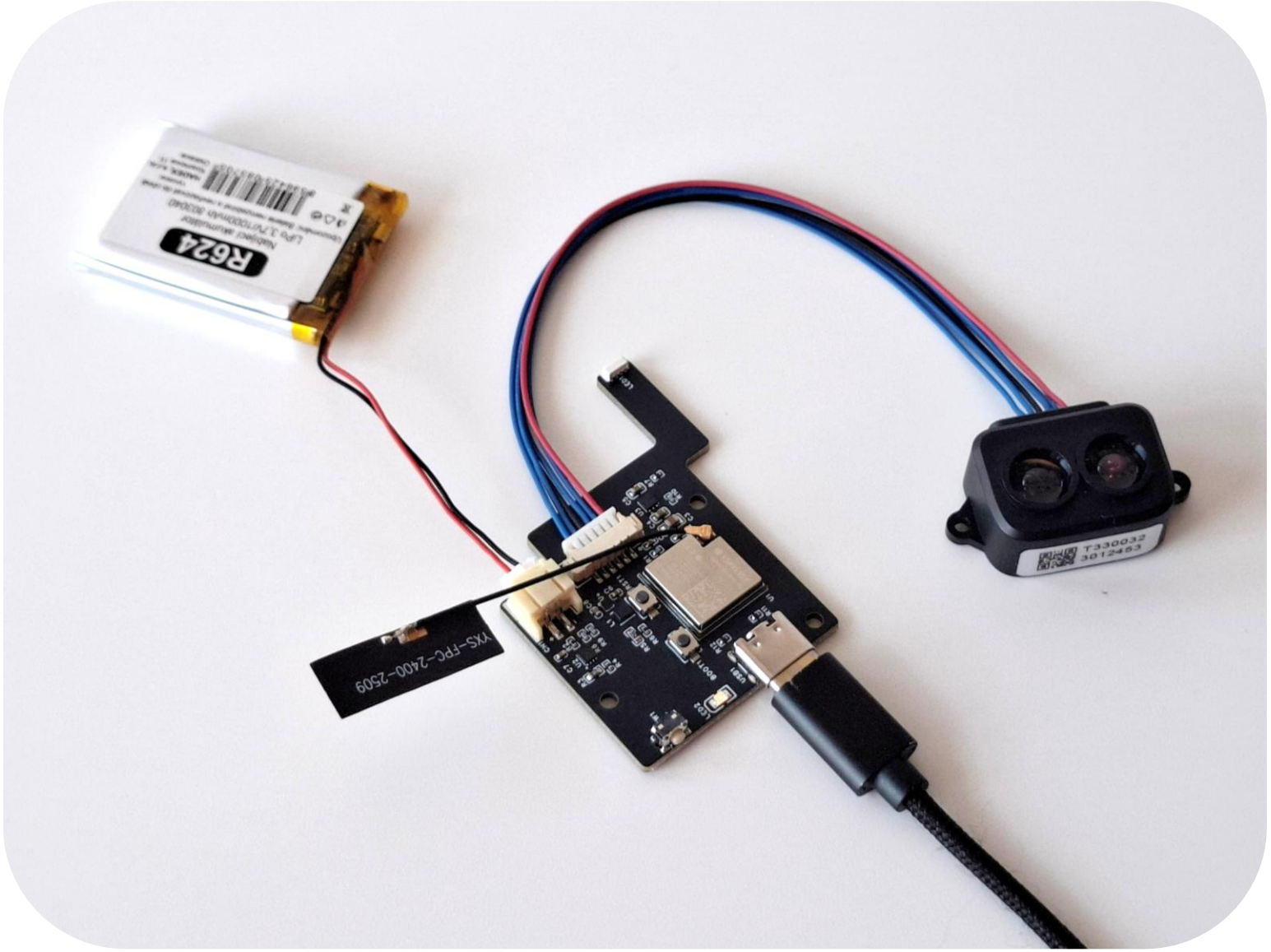
⚠ Caution: The polarity must match. If it does not, swap the polarity on the battery connector.

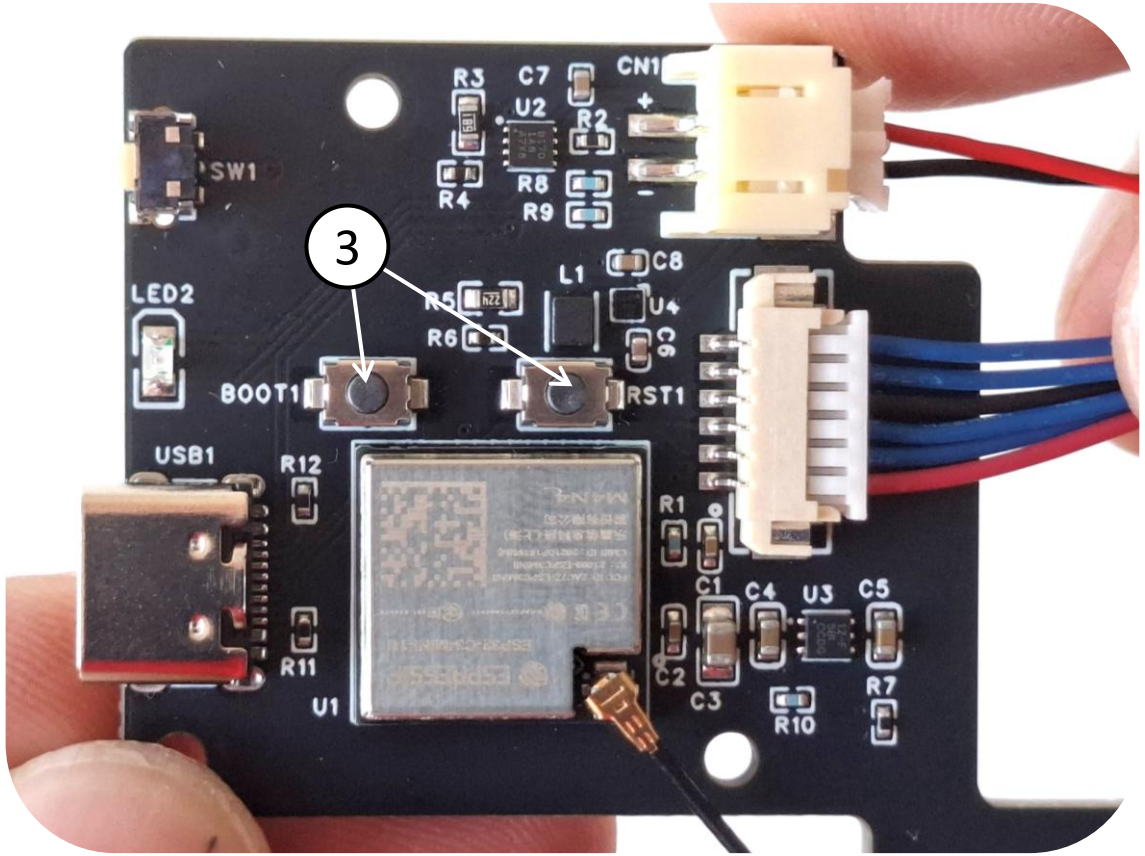


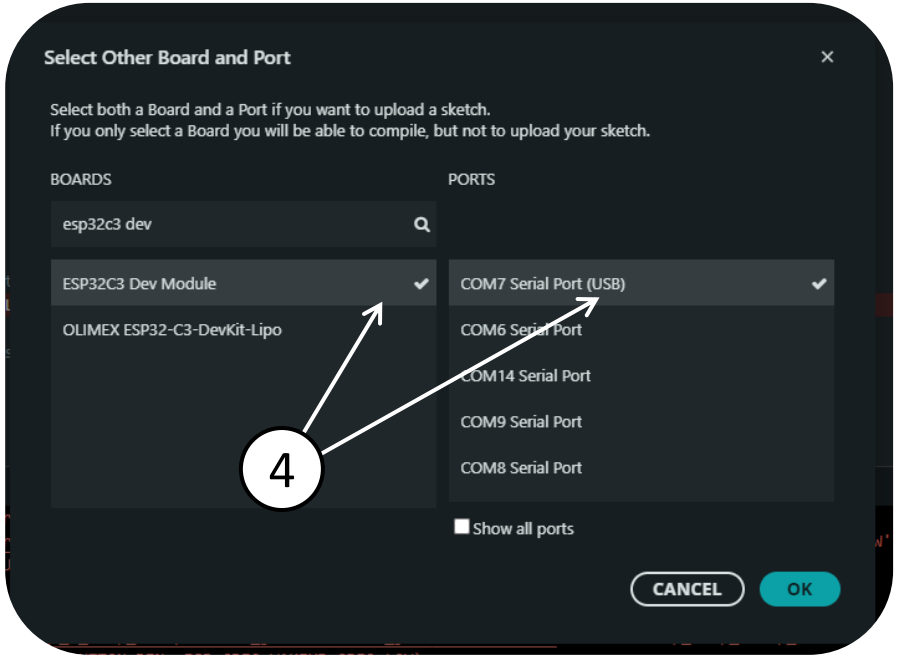
Let's upload
the firmware

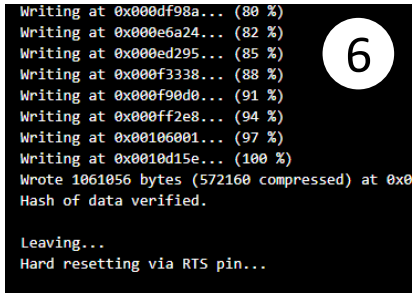
Before you start:

- Ensure all parts are connected (especially the battery) – without it, the PCB will not function!
- **Requirements:**
  You will need **Arduino IDE** installed on your PC.

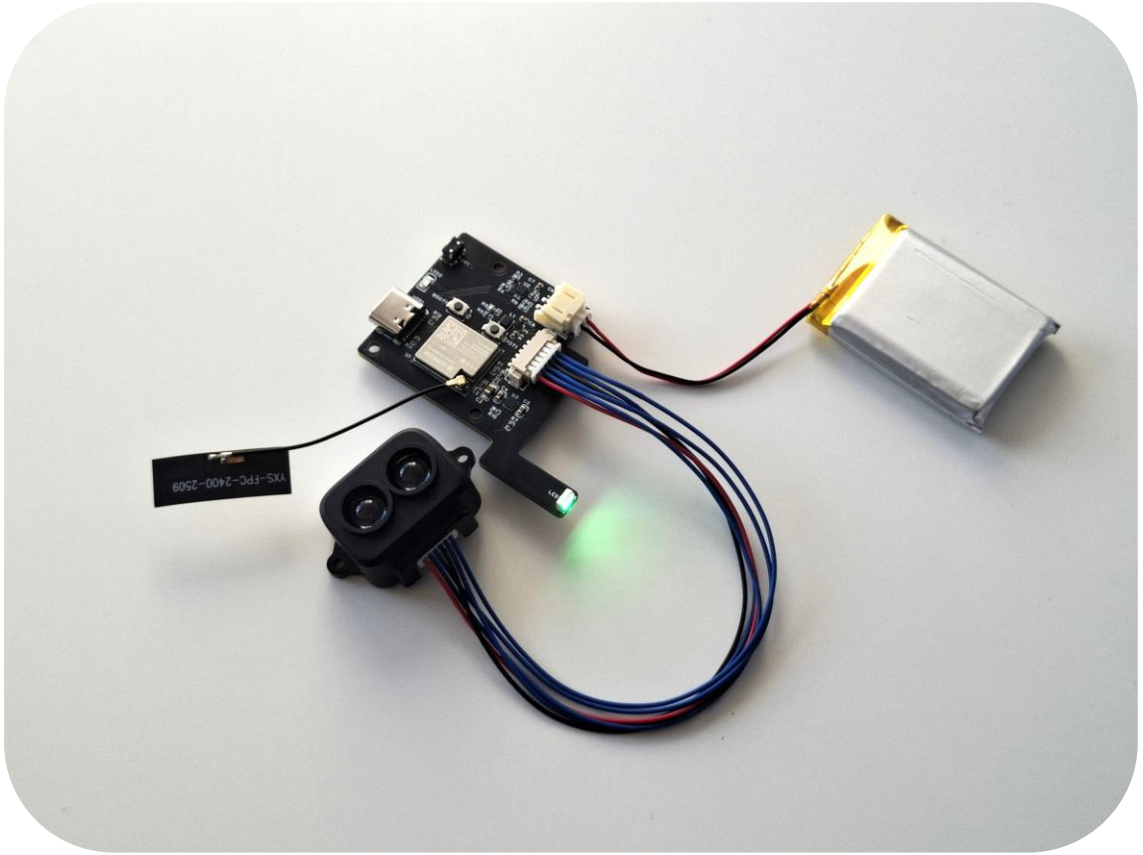


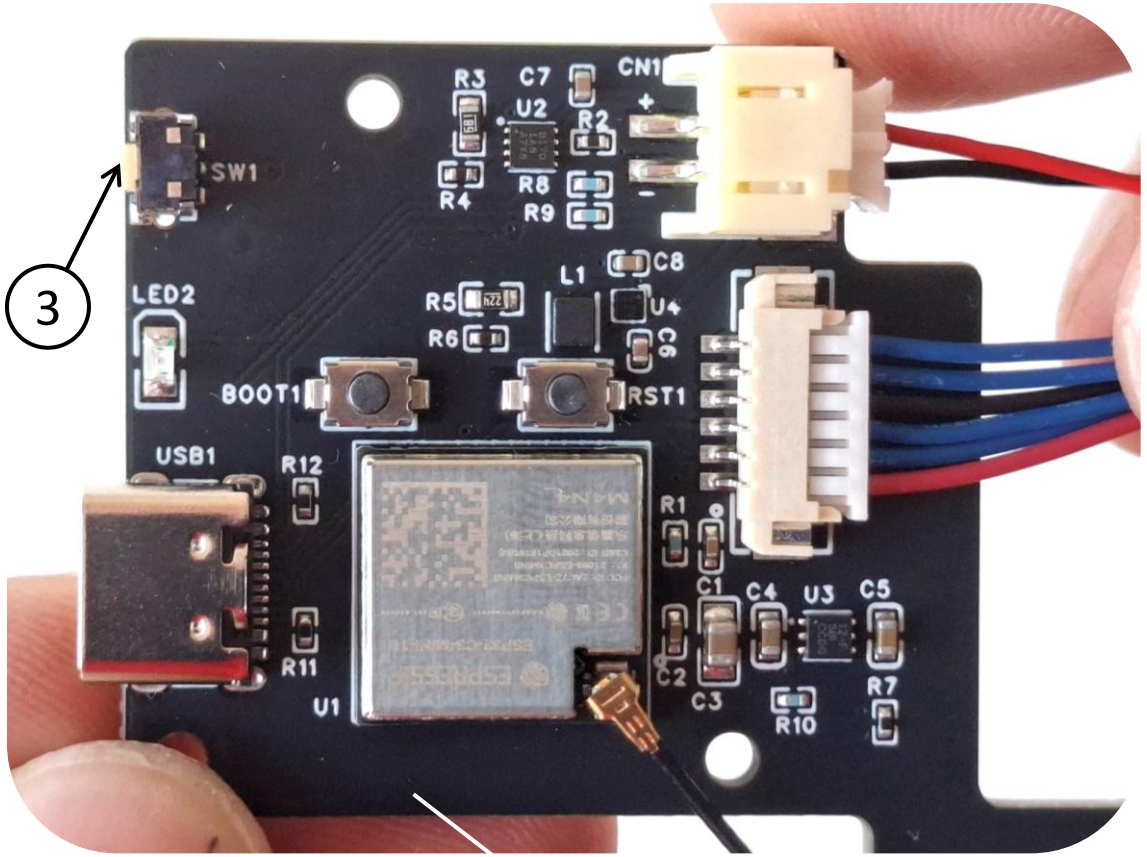






1. Open the **Arduino IDE** and open the firmware [file]. Install any required libraries if necessary.
2. Connect the board to your PC via USB-C.
3. Press and hold the **BOOT1** button, then press the **RST1** button. Afterwards, release the **BOOT1** button.
4. In the software, set the board to **ESP32C3 Dev Module** and select the appropriate port.
5. Press the **RST1** button again. ⚠️ The board will disconnect and reconnect, but leave it in this state. ⚠️
6. Upload the program to the board. Upon successful upload, you will see the following in the console





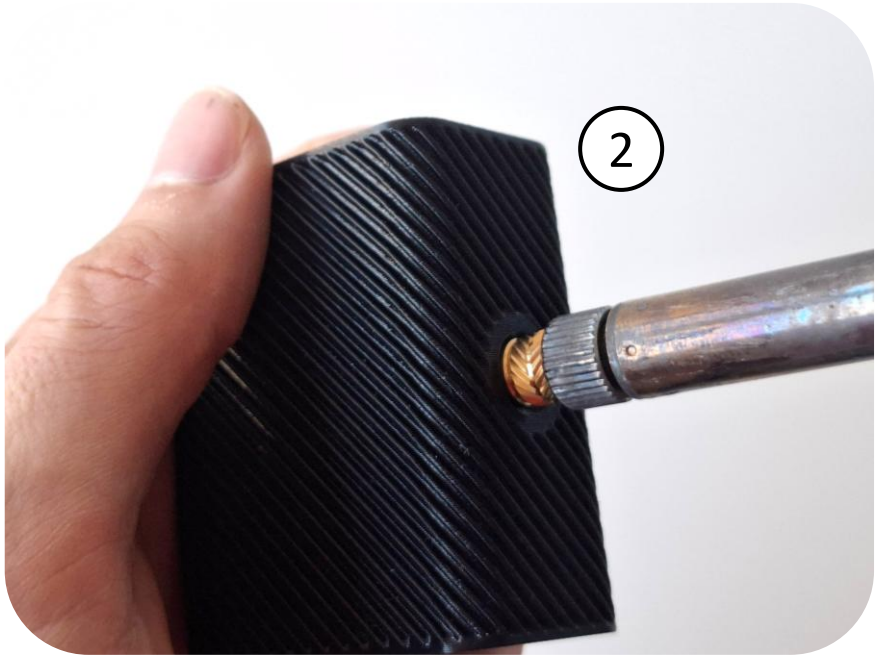
Check the following after uploading and disconnecting from USB-C:

1. The LED light should glow green.
2. If you wave your hand in front of the laser, it should turn red.
3. Press and hold the **SW1** button for 2 seconds to put the board into deep sleep mode.
4. Press and hold it again for 2 seconds to wake the board back up.



Let's put it into the enclosure

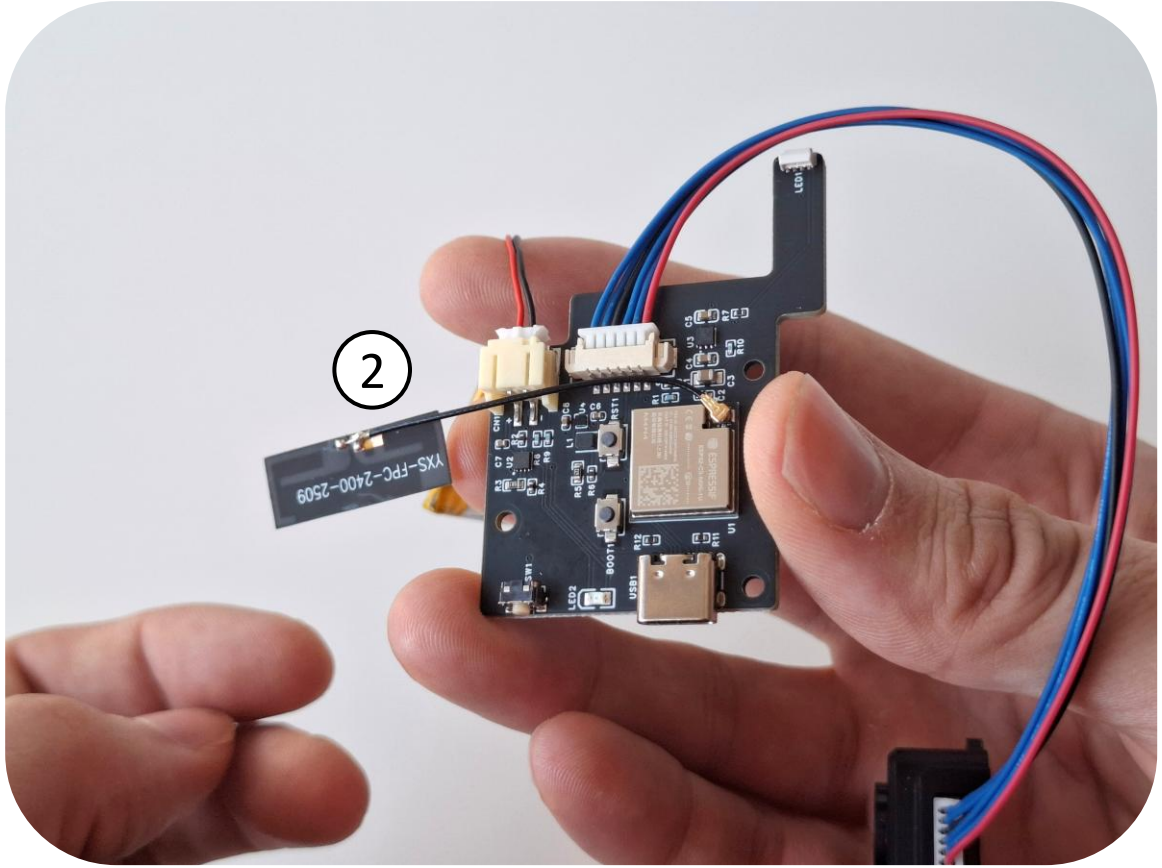




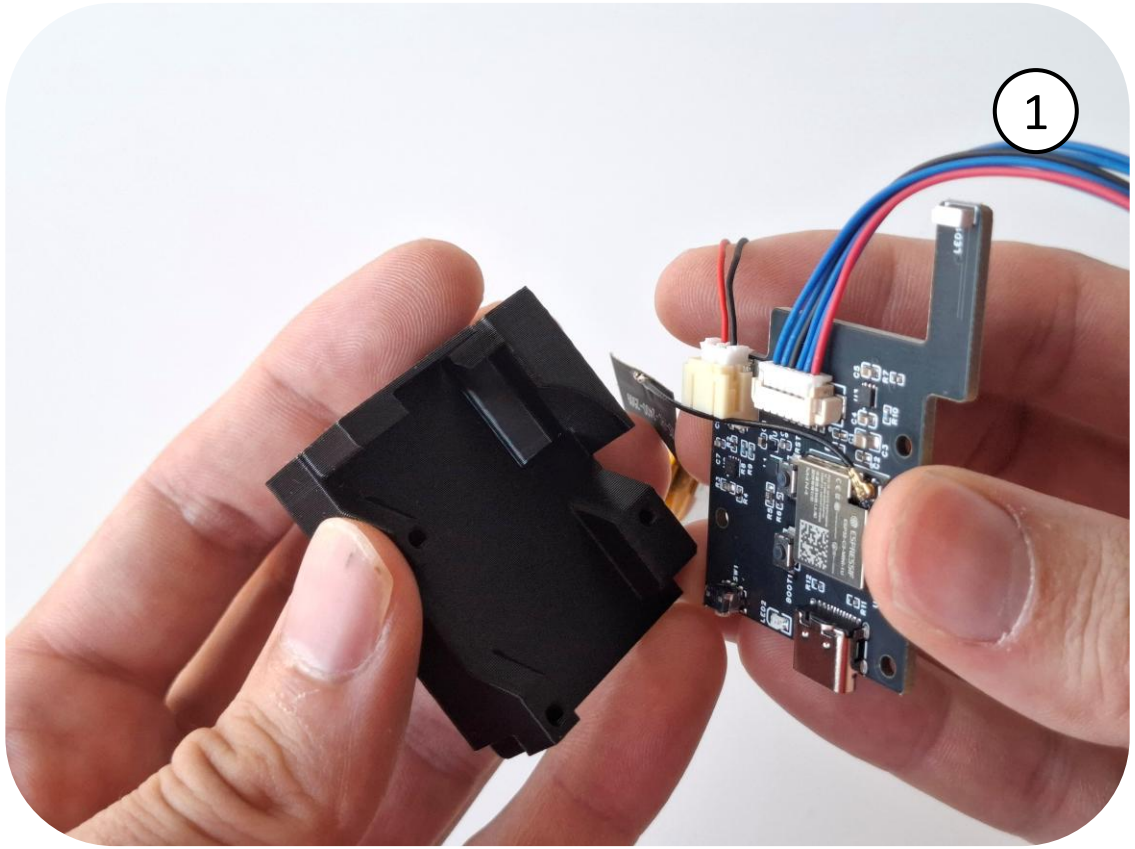


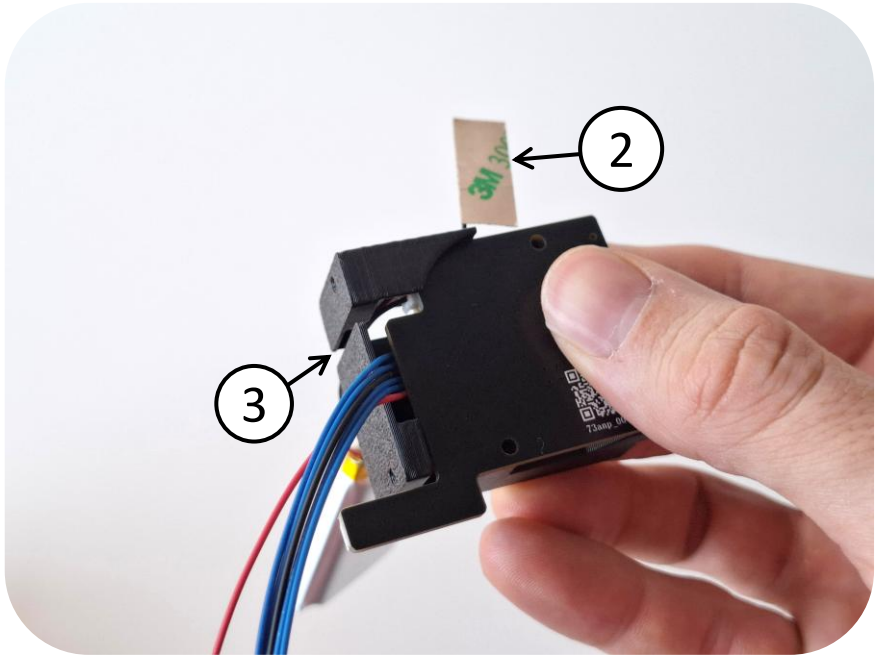
1. Using a soldering iron, heat and carefully press the threaded insert into the hole.
2. The threaded insert should not protrude above the surface.

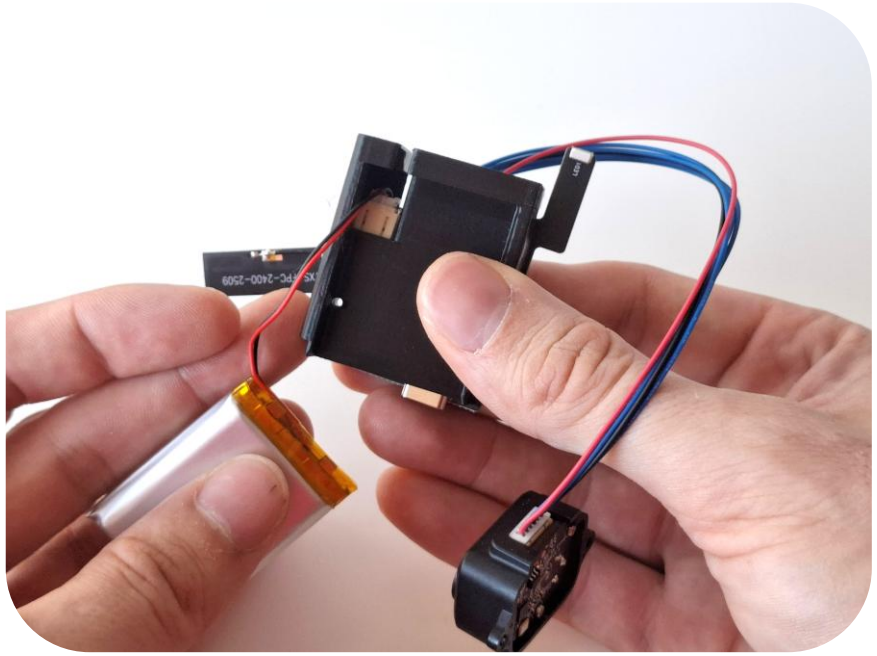




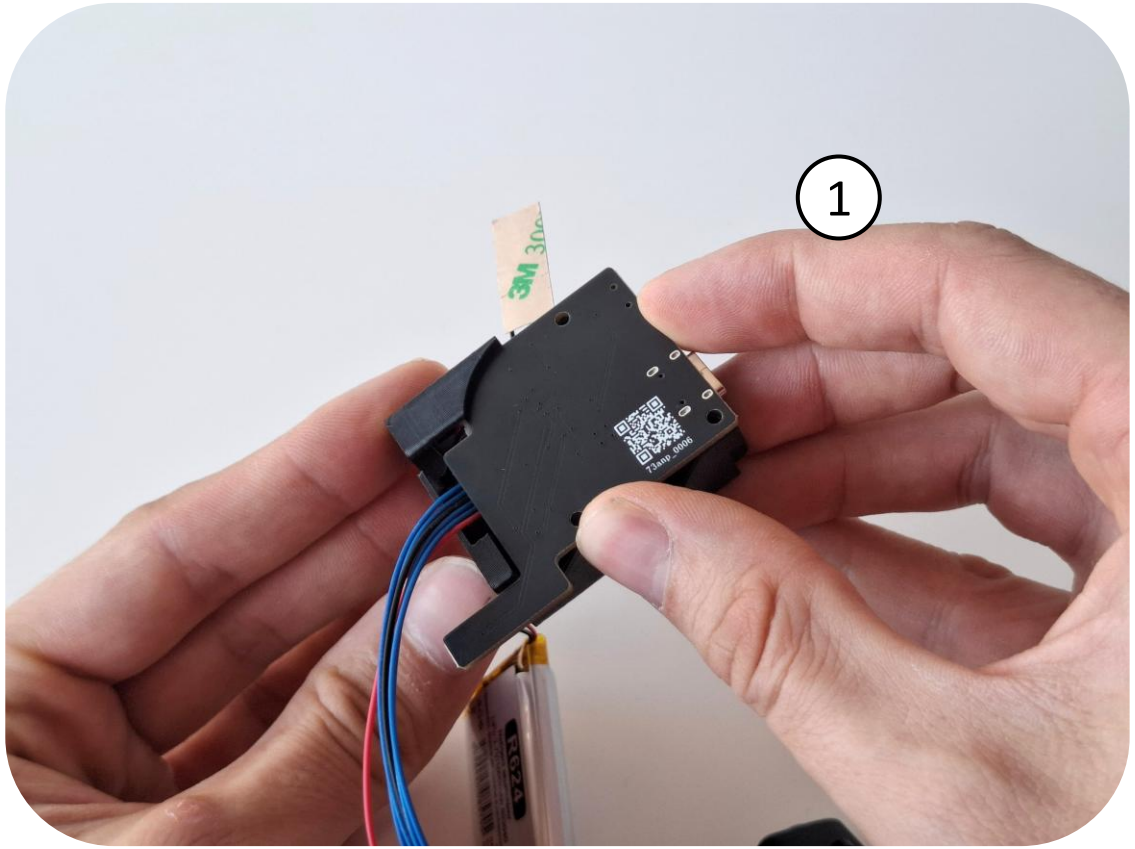
1. Prepare the internal cartridge.
2. Adjust the position of the antenna as shown in the picture.

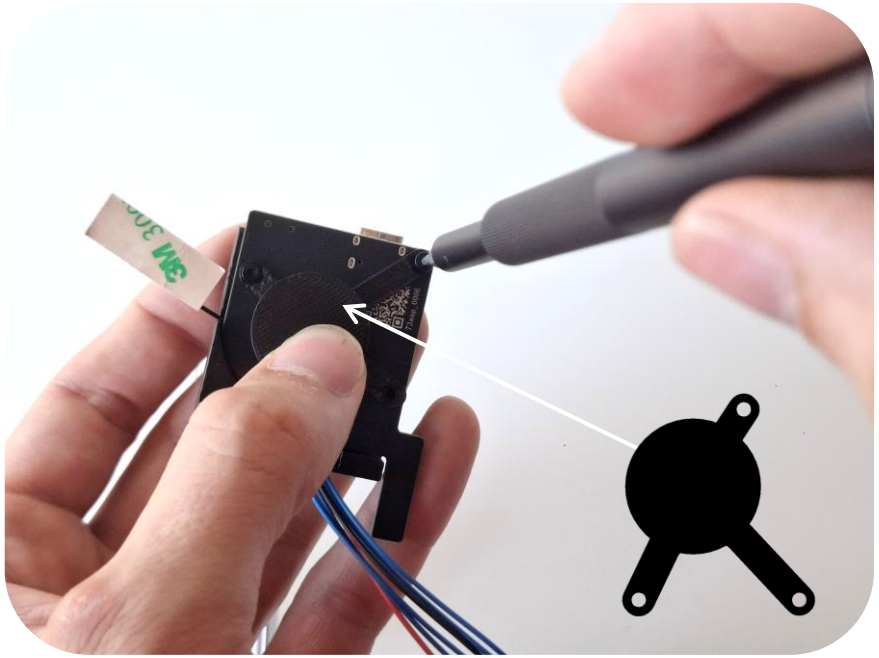


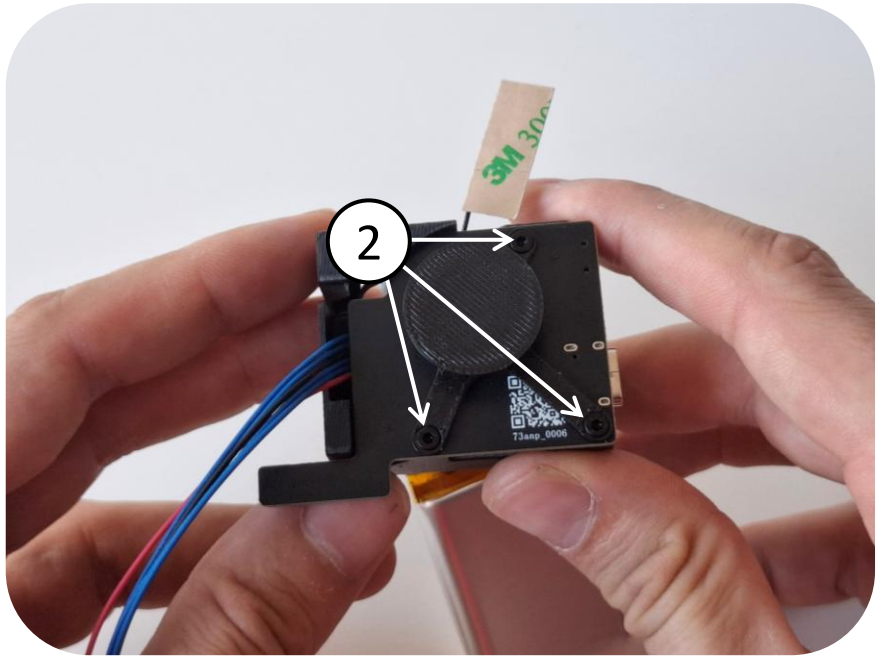




1. Place the PCB onto the cartridge as shown in the picture.
2. The antenna should protrude through the slot.
3. Route the battery cable through the groove to the other side.

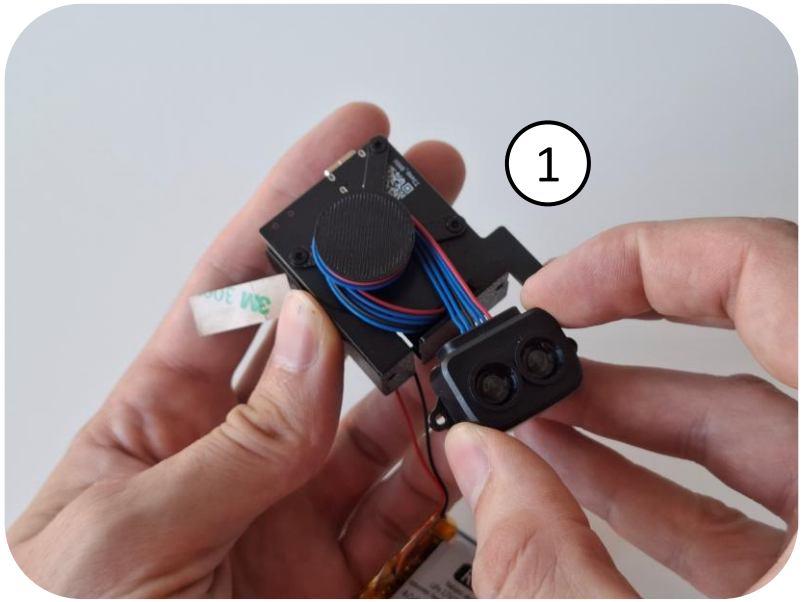


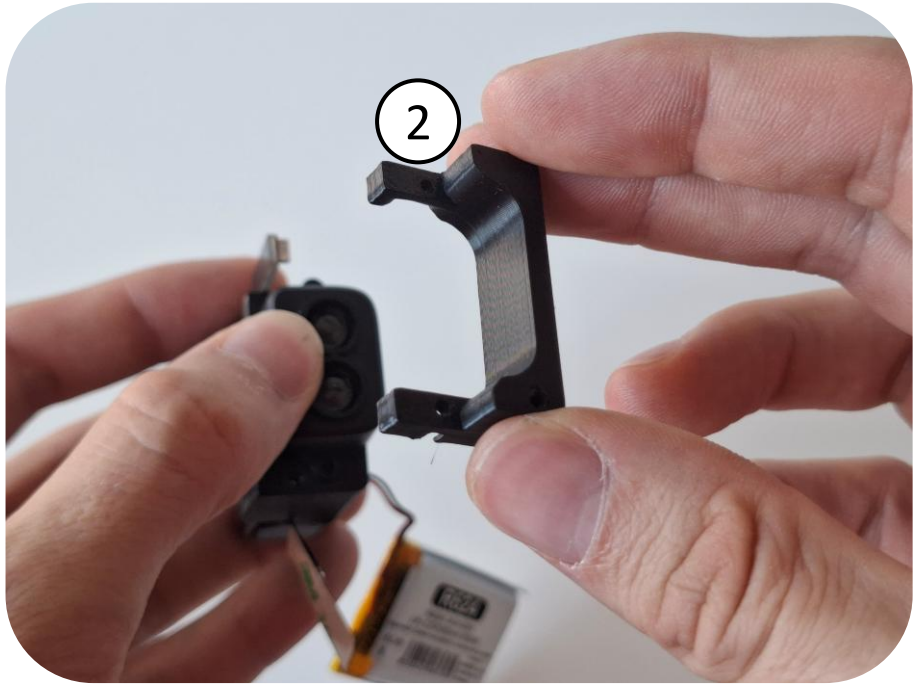


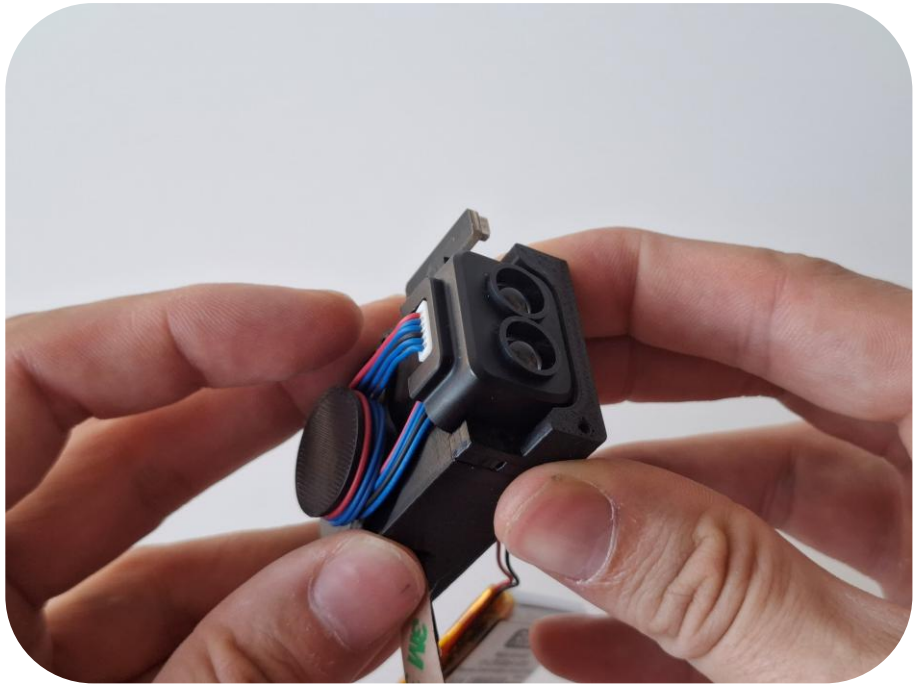


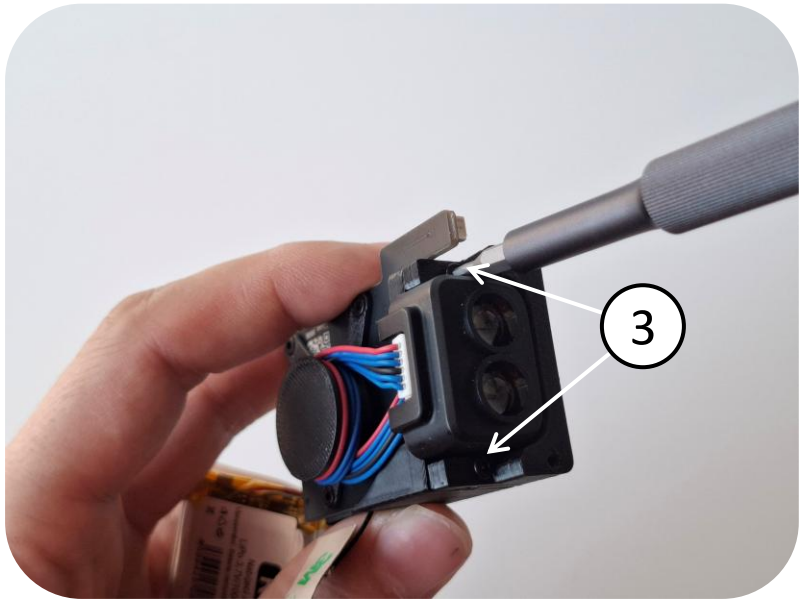
1. Align the PCB as shown in the picture.
2. Secure the fixture to the PCB and cartridge using 3 screws.

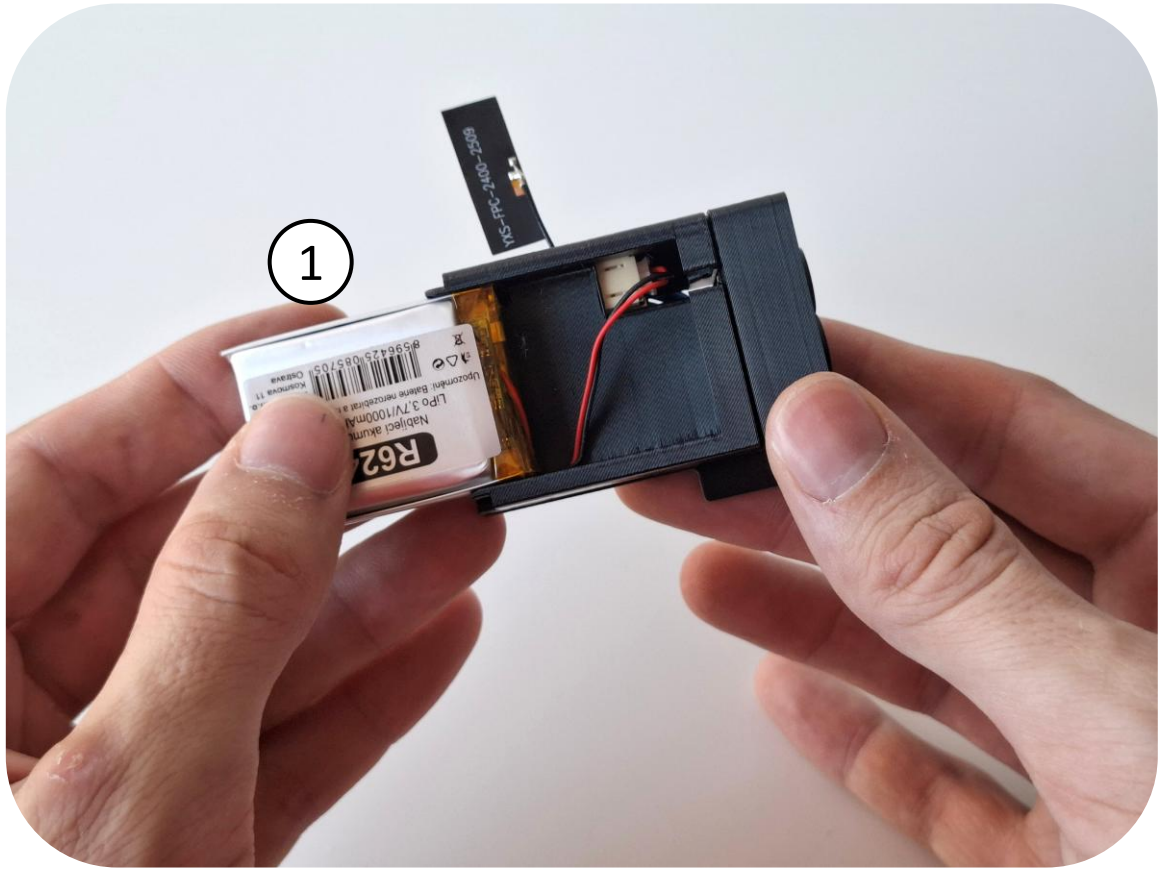
1. Wrap the cables from the LiDAR sensor (laser) twice as shown in the photo.
2. Place the bracket onto the laser.
3. Tighten the 2 screws.









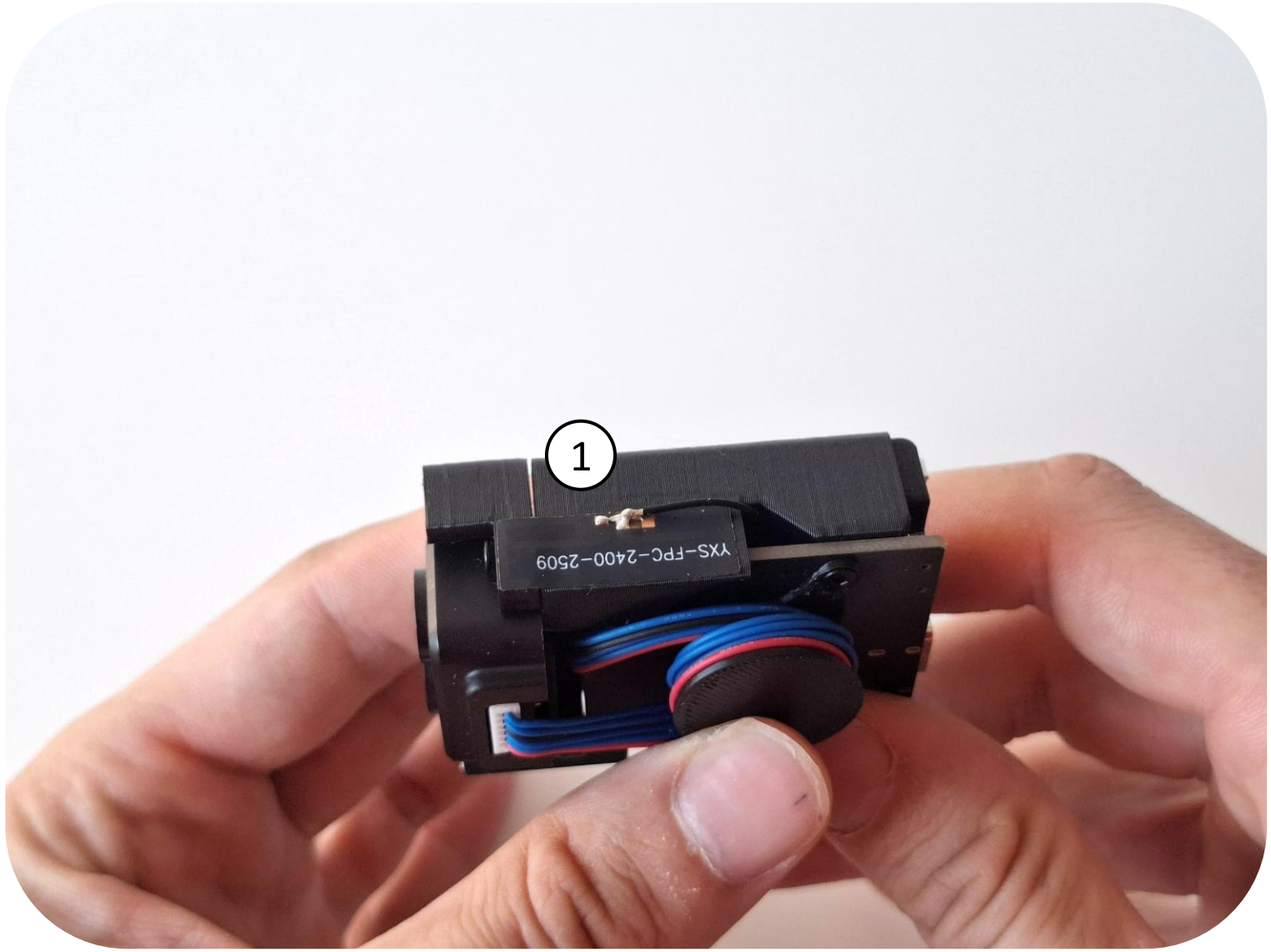


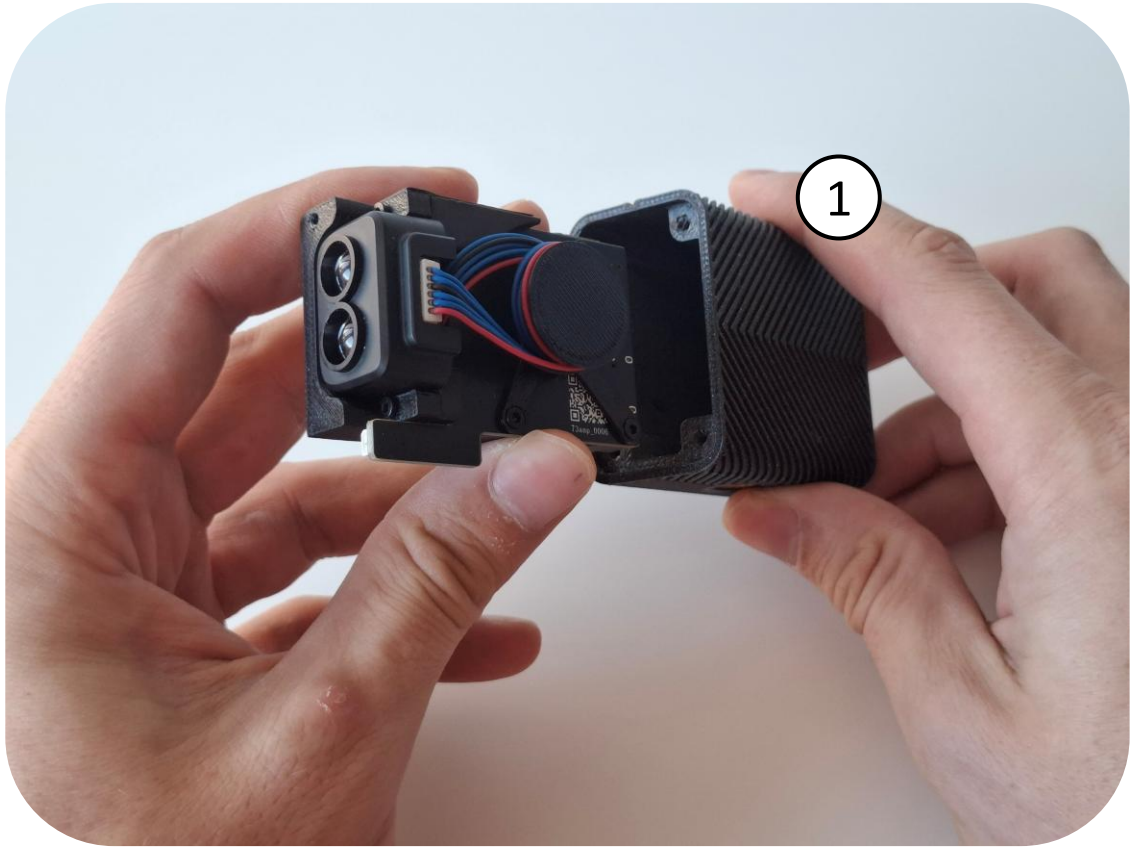


1 Insert the battery and cables into the slot as shown in the photo.

⚠️ Make sure the cables do not protrude or get pinched. ⚠️

1 Stick the antenna to the top side of the cartridge.

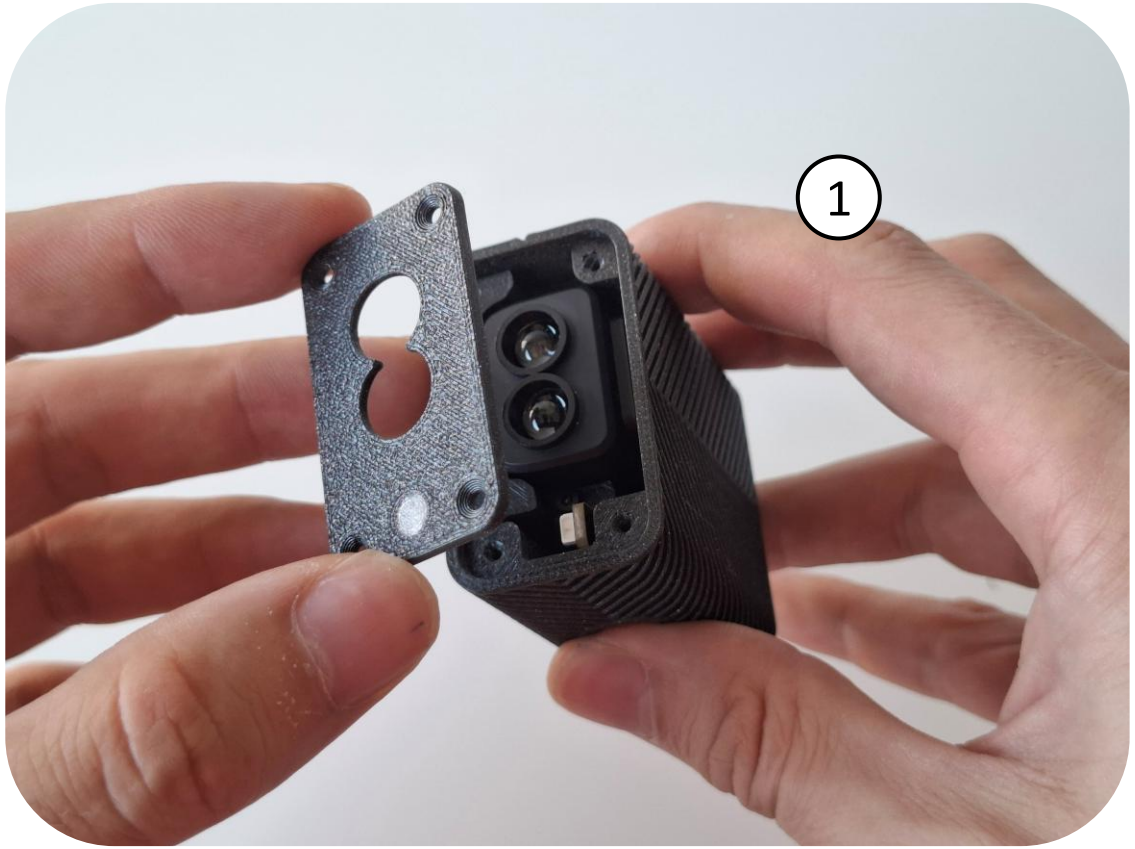








1. Carefully insert the cartridge into the enclosure.
2. Ensure the USB-C port aligns properly with the pass-through opening.







1. Attach the front cover.
2. Tighten the 4 screws.

Ready for RUN!

